GEOINT CASE

Sentinel Copper Mine (Zambia)

Supplier Reliability and Infrastructure Risk Assessment

Zambia

Subject	Sentinel Copper Mine
Jurisdiction	Zambia
Case type	Supplier reliability / infrastructure risk
Imagery	Satellite and aerial imagery embedded in the source
Source basis	Satellite imagery and public sources

RESEARCH QUESTION

Preliminary spatial assessment of the operational footprint, supporting infrastructure, and functional zoning to evaluate the scale and structure of a large industrial mining operation.

Prepared from public sources only. No private, paid or non-public data was used.

Objective: Preliminary spatial assessment of the operational footprint, supporting infrastructure, and functional zoning to evaluate the scale and structure of a large industrial mining operation.

Image 1 - Functional Area Overview

Red - Total area of interest The red boundary represents the overall area under assessment, covering approximately 470 km2 . Blue - Reservoirs Two reservoirs are identified:

- Reservoir 1
- Area: 29 km2
- Includes associated infrastructure
- Reservoir 2
- Area: 12 km2
- Also supported by operational infrastructure Both reservoirs were previously forested areas and were artificially flooded during the development of the mining operations starting from 2012 . Green - Residential area

Four distinct residential zones are identified:

- Area 1
- Low-density development
- Approximate area: 2 km2
- Limited supporting residential infrastructure
- Average house size: ~25 m2
- Based on indirect indicators, appears to be low-income housing
- Area 2
- Low-density development
- Approximate area: 2 km2
- Presence of necessary residential infrastructure
- Higher construction quality compared to Area 1
- Average house size: ~62 m2
- Area 3
- Structured layout with organized planning
- Medium-density, standardized development
- Includes both private houses and dormitories
- Private houses: ~70 m2, two floors
- Dormitories: ~170 m2, single floor
- Supporting residential infrastructure present
- Area 4
- Most developed residential zone
- Located near Reservoir 1
- Detached private two-story houses with individual plots
- Average house size: ~280 m2
- Well-developed urban infrastructure, the most advanced among all residential areas Orange - TSF The TSF is located adjacent to Reservoir 1 on its eastern boundary. A dam structure connects the reservoir to the facility, indicating a water supply function.

Visual indicators suggest active use . Pink - Industrial mining areas Three industrial mining zones are identified:

- Area 1 (Primary)
- Approximate area: 50 km2
- Contains full operational infrastructure

- Includes the largest open pit
- Area 2
- Approximate area: 8 km2
- Contains basic operational infrastructure
- Area 3
- Approximate area: 0.7 km2
- No visible infrastructure other than the open pit itself Preliminary Operational Interpretation The spatial structure indicates a large-scale, integrated industrial mining system designed for long-term operation. The presence of multiple industrial mining areas, extensive TSF

infrastructure, engineered reservoirs, and tiered residential zones suggests a high level of capital investment and operational maturity. The differentiated residential areas, including planned housing and dormitory facilities, indicate a structured workforce model consistent with a remote, self-contained operation. The scale and configuration of water infrastructure, including artificially created reservoirs connected to the TSF, further support continuous large-volume processing capability. Overall, the site layout reflects a centralized industrial complex with dedicated operational, environmental, and workforce support systems. Such a configuration is typical of high-throughput mining assets where production continuity depends primarily on the stability of critical external infrastructure and logistics rather than on local resource or settlement constraints.

Logistics and Access

Access Routes (Images 2-6)

The site can be accessed via three primary entry routes. 1. Main entrance Coordinates: 12°10'19.21"S, 25°32'26.75"E (Image 3) Access is provided from a national highway via a two-lane paved road. This road serves as the primary transport corridor and includes multiple branches connecting to residential areas and the airport. Before entering the industrial mining zone, a controlled access point is located at: 12°15'16.33"S, 25°25'18.93"E (Image 5) The presence of a checkpoint indicates controlled access to the core operational area. A dedicated turnoff toward the airport is also present along the main route. 2. Secondary entrance Coordinates: 12°06'05.63"S, 25°29'30.65"E (Image 4) This access point is located north of the main entrance and passes through a residential area. The road is unpaved and includes several local branches. No visible checkpoint or controlled access infrastructure is observed. This route appears to serve secondary or local traffic rather than heavy industrial transport. 3. Airport access Coordinates: 12°15'04.77"S, 25°26'15.88"E (Image 6) The airport includes a single runway with the following dimensions:

- Length: approximately 2,700 meters
- Width: approximately 35 meters The facility appears designed for medium-size aircraft and supports personnel transport and operational logistics.

Preliminary Logistics Interpretation

The logistics structure indicates a primary dependence on a single paved road connected to the national highway , which functions as the main heavy-transport corridor for the operation. The secondary unpaved route does not appear suitable for large-scale industrial transport and likely serves local or limited operational access. The airport provides dedicated air access, supporting personnel rotation and operational mobility. The runway length is sufficient for medium-capacity aircraft commonly used in remote industrial operations. Overall, the site logistics model reflects a centralized access system with limited redundancy for heavy ground transport , suggesting potential exposure to disruption affecting the primary road corridor.

Power Supply

IMAGE 7

The entire Sentinel Copper Mine is supplied through a single primary electrical intake facility located at 12°17'27.83"S 25°17'34.88"E (Image 7). The power infrastructure area covers approximately 33,000 m2 , indicating a large-scale high-voltage substation designed to support energy-intensive mining and processing operations. The facility appears to receive electricity via two double-circuit high-voltage transmission lines mounted on tower structures. These lines run within a single transmission corridor extending beyond the operational area, with no evidence of geographically independent alternative supply routes. Based on the physical scale of the substation and transmission configuration, the installation likely operates at high-voltage class (estimated ≥220 kV) and functions as the central power intake and distribution node for the entire mining complex. A smaller internal substation located within the industrial zone appears to be fed from this primary node, indicating a hierarchical internal distribution structure. The observed configuration suggests a high level of dependency on a single external power corridor and a single intake facility . Although multiple circuits are present, their co-location within the same corridor limits system-level redundancy. Any disruption affecting the transmission corridor or the primary substation would likely result in a large-scale loss of external power supply and significant operational impact across the site. Given the continuous and energy-intensive nature of open-pit copper mining and processing operations, stable grid connectivity represents a critical operational requirement for the Sentinel Mine.

Processing Capacity

Images 8, 9, 10 The Sentinel Copper Mine operates a centralized processing model, where ore from multiple extraction areas is transported to a single industrial processing complex. The spatial structure indicates a unified, industrial-scale material flow system. Industrial Zone 1 (Primary Production Hub) - Image 8 Industrial Zone 1 represents the core operational area of the entire facility. Processing infrastructure

- The only processing zone on site is located here.
- Total area: approximately 2.5 km2 .
- The facility serves all extraction areas across the project.
- Pipeline infrastructure connects the processing plant directly to the TSF .
- No tailings pipelines originate from other industrial zones.
- Several water reservoirs are located within the processing area, indicating integrated water management for continuous operations. Extraction
- Extraction Area 1 : ~ 3.6 km2 .
- Exact depth and volume cannot be determined from imagery.
- However, multiple indicators confirm active mining:
- extensive haul road network
- fresh bench development

- large exposed surfaces
- Open-source information suggests significant pit depth, consistent with large-scale open-pit operations. Waste handling
- Waste Handling Areas 1 and 2 : combined area ~ 15 km2 .
- Terraced structure and ongoing expansion indicate active use and high stripping volumes. Material flow assessment
- Analysis of road connectivity indicates that ore from other extraction areas is transported to this zone for processing.
- This confirms Industrial Zone 1 as the central processing hub for the entire mining operation. Industrial Zone 2 (Secondary Extraction Area) - Image 9

This zone functions as a supplementary production area. Extraction

- Extraction Area 2 : ~ 0.85 km2 . Waste handling
- Waste Handling Areas 3 and 4 : combined area ~ 1.5 km2 . No processing infrastructure or tailings pipeline connections are observed. Material is transported to the primary processing facility in Industrial Zone 1. Industrial Zone 3 (Inactive / Legacy Area) - Image 10

This zone shows signs of limited or discontinued activity. Extraction

- Extraction Area 3 : ~ 0.15 km2 .
- The pit is flooded , indicating cessation of active mining. Waste handling
- Waste Handling Area 5 : ~ 0.12 km2 . No active operational infrastructure is visible. Capacity Interpretation

The following structural indicators support an industrial-scale processing capacity :

- Centralized processing facility (~2.5 km2)
- Multiple extraction areas feeding a single plant
- Extensive waste rock disposal (~16+ km2 total)
- Dedicated tailings pipeline system connected to a large TSF
- Integrated on-site water storage This configuration is consistent with a high-throughput, continuous copper production operation rather than pilot-scale or phased development.

Final Assessment and Operational Risk Summary

1. Operational Scale

The Sentinel Copper Mine demonstrates clear characteristics of a large-scale industrial operation. Key indicators include:

- Multiple active open-pit mining areas
- A centralized processing complex serving all extraction zones
- Extensive waste handling infrastructure with large engineered dumps
- A large tailings storage facility (TSF) integrated into the operational water cycle
- Dedicated industrial water reservoirs
- Permanent residential zones and on-site airport infrastructure The spatial footprint, infrastructure intensity, and level of engineering indicate a major industrial producer rather than a pilot or developing project .

2. Operational Maturity

The site shows strong indicators of long-term operational maturity:

- Multiple extraction areas at different stages of development
- Large accumulated waste volumes indicating multi-year production
- Engineered reservoirs created through controlled flooding of former forest areas

- Structured workforce accommodation ranging from basic housing to high-standard residential zones
- Dedicated airport supporting regular personnel and logistics movement Overall, the infrastructure profile is consistent with a fully developed, long-life mining operation designed for continuous industrial production.

3. Critical Infrastructure Dependencies

Despite its scale, the operation relies on several critical infrastructure elements with limited redundancy. Power Supply

- The entire operation is supplied through a single primary energy node (~33,000 m2).
- Power is delivered via a single high-voltage transmission corridor.
- Internal distribution networks originate from this node. This configuration represents a single-point-of-failure risk for the entire production system. Processing
- All extracted material from multiple mining areas is transported to a single processing zone (~2.5 km2).
- No secondary processing facilities were identified.
- Processing output is directly linked to the TSF via pipeline infrastructure. Any disruption to this facility would directly halt production.

Water System

- Operations depend on large artificial reservoirs (29 km2 and 12 km2).
- Reservoirs support processing and tailings water management.
- The TSF is hydraulically integrated with Reservoir 1. This indicates high dependence on engineered water control systems . Site Access and Logistics
- Primary access is via a single national highway connection.
- Controlled entry through a main checkpoint.
- Secondary access exists but is unpaved and less controlled.
- On-site airport provides operational flexibility but has a single runway. The operation is therefore dependent on

limited ground access infrastructure .

4. Key Operational Risks

The main operational vulnerabilities identified are:

- Power disruption risk Single transmission corridor and primary substation.
- Processing bottleneck risk Entire production dependent on one processing facility.
- Water system risk Reliance on large engineered reservoirs and TSF integrity.
- Logistics disruption risk Dependence on a single highway connection for heavy transport.
- Infrastructure concentration risk Core operational functions are highly centralized.

5. Supplier Reliability Assessment

The Sentinel Copper Mine demonstrates the spatial and infrastructural characteristics of a stable, large-scale industrial producer capable of supporting continuous commercial output. However, operational continuity is highly dependent on the integrity of several critical infrastructure elements. The most significant vulnerabilities relate to:

- Single-source power supply
- Centralized processing capacity
- Engineered water system stability
- Limited ground logistics redundancy Conclusion: The operation can be assessed as a high-capacity and operationally mature supplier , but with moderate infrastructure concentration risk due to multiple single-point dependencies.

Executive Summary

Sentinel Copper Mine - Operational Reliability and Supply Risk Assessment Objective This assessment evaluates whether the Sentinel Copper Mine represents a stable large-scale copper supplier and identifies key infrastructure or operational factors that may affect production continuity. Operational Scale and Status Geospatial analysis indicates that Sentinel is a large, fully operational industrial mining complex . Indicators include:

- Multiple active open-pit extraction areas
- A centralized processing facility serving all mining zones
- Extensive waste handling areas (~15 km2)
- A large tailings storage facility (TSF) integrated with the site water system
- Dedicated industrial reservoirs (29 km2 and 12 km2)
- Permanent workforce residential zones of varying standards
- On-site airport with a 2,700 m runway The scale and infrastructure footprint confirm industrial-level production capability and long-term operational maturity . Production Model

The operation follows a centralized processing model :

- All ore from multiple pits is transported to a single processing zone (~2.5 km2)
- Processed material is hydraulically transferred to the TSF
- No secondary processing facilities were identified This structure supports high-volume operations but creates a processing concentration risk . Infrastructure Dependencies

Power

- Entire operation supplied through a single primary substation (~33,000 m2)
- Connected via a single high-voltage transmission corridor Water
- Processing and tailings operations dependent on large engineered reservoirs
- TSF hydraulically connected to Reservoir 1 Logistics
- Primary ground access via a single national highway connection
- Controlled main checkpoint
- Secondary unpaved access route
- On-site airport provides personnel and light logistics support Key Risk Factors The main potential disruption risks are: 1.

Single-point power dependency Failure of the transmission corridor or primary substation would affect the entire operation. 2. Processing bottleneck All production depends on one processing facility. 3. Water system dependency Operations rely on engineered reservoirs and TSF stability. 4. Limited ground logistics redundancy Heavy transport dependent on a single highway connection. 5. Operational centralization Core functions (power, processing, water) are geographically concentrated. Supplier Reliability Assessment Sentinel Copper Mine demonstrates characteristics of a high-capacity, mature industrial producer capable of sustained commercial output. However, operational continuity is sensitive to failures affecting:

- External power supply
- Central processing infrastructure
- Water management systems
- Primary transport access Overall Assessment: High production capability with moderate operational vulnerability due to infrastructure concentration and limited redundancy.